
Sensibility analysis of IoT calibration via ML methods for NH₃

PENNANECH Hugo - Università di Catania - Facoltà di Agraria - Dipartimento DI3A

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1 Introduction

This document aims at presenting the elements to take into account during the process of calibration of NH3 concentrations of a low cost/low precision instrument on a high cost/high precision instrument. The study of the sensitivity of the calibration is indeed mandatory in order to be able to deploy to a large scale the low cost instruments with a good precision. To do so ,measurement campaigns have been done in order to collect data.

1.1 Description of the instruments

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1.2 On site situation

The calibration work will be done for 2 campaigns : June 2025 (12-25), September 2025 (11-24). Each time, a low quality sensor (rabbit) is located next to the high precision one (innova). The sample locations (SL) are positioned as followed:

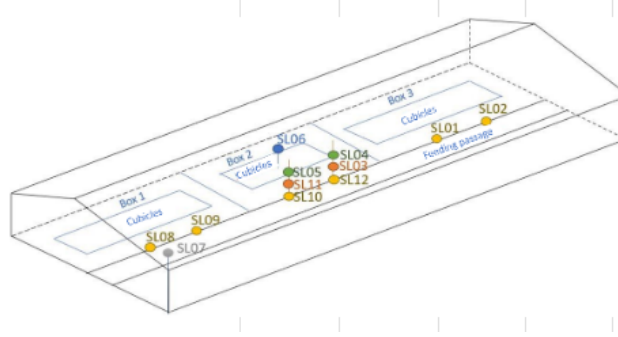


Figure 1: Sample location inside the famr

On those campaigns, the following couples were measured:

IDs	June				September			
SL (innova)	3	11	5	4	11	4	5	3
Rabbit	1	2	4	6	2	3	4	6

Table 1: Mapping ID of low cost-high cost instrument

1.3 Dataset description

As the sampling times of the instruments are not the same, the data were aligned temporally on the longest sampling time. On those campaigns the delta T between two measurements is of 12mn for the June campaign while it was of 10mn in September.

With the innova instruments we were able to collect the concentration of NH3, CO2, CH4 while with the rabbit instruments we were able to collect NH3, CO2, temperature, humidity and THI. As the main focus of this article is NH3 we kept as features:

- Innova (target column): NH3
- Rabbit (features): it was decided to kept all the features for the training.
- Spatio temporal location (grouping columns/ labels) : datetime, id rabbit, id channel (SL), height of the sensor

The dataframe built was finnaly composed of more than 8000 lines.

2 Methodology: sensibility tests

The following method has been implemented and will be considered as the reference protocol on all the study (if not specified otherwise).

- Definition of :
 - Y = target col (NH3 concentration from innova)
 - X = training features (NH3 concentration, CO2 concentration ,temperature, humidity, THi from rabbit instruments)
- Selection of one model with the associated parameters grid
- Split of X and Y for training and test (respectively 70% and 30% with ShuffleSplit)
- Application of a scaler on X and Y test/train (Python Standard Scaler ($f : x \mapsto (x - \text{mean}(X)) / \text{std}(x)$))
- Cross validation on the parameters grid with X_{train} and Y_{train} to find the best parameters for the model (the one maximising R^2)
- Re training of the model with the best parameters on all Xtrain and evaluation of the R^2

2.1 First tests

A baseline test has been performed on various models in order to choose the models the more adapted to our case. The models tested are the following: Linear, Ridge MLP, XGB, RFR, CatBoost, LGBM. The rest of the paper will be then done with the best performing models.

2.2 Hyper-parameters sensibility and reproductibility

2.2.1 Hyper-parameters sensibility

To check deeply the impact of the parameterization of those models, an intelligent search over hyperparameters is chosen (called Optuna). In inputs are given only a range for each parameters instead of fixed values as well as a number of test to be done and the solver converges automatically towards the best one. The differences between the best model (best R^2 test) and the worst one found during will be studied.

To be sure that the model generalizes at best the situation (and do not overfit), another test will be done incrementing a penalized score equals to the $R^2(\text{test CV}) - \lambda \text{ gap}(R^2(\text{train CV}), R^2(\text{test CV}))$ if $R^2(\text{train CV})/R^2(\text{test CV}) > 1.05$ else only $R^2(\text{test CV})$. Lambda will be taken as 5.

The results will be then compared regarding the direct results but also on the reproductibility of those last ones.

2.2.2 Reproducibility of the results

Choosing how to choose the data to train a model is a crucial element to take into account. The general method used in this paper is the ShuffleSplit to separate the training dataset from the test one (randomly choosed). However one can assume that this method is not applicable or not well suited for real measurement. Indeed the model trains on all the period meaning that he sees all the different regimes of conditions. This model might be seen as 'perfect' or maybe fully trained on all regimes possible. A model trained on some regimes that encounters new values might not perform as well. Consequently in this section , an initial TimeSerie split (meaning that the 70% oldest data will be used for training and the 30% the more recent will be use for test) will be used to compare the results of the Optuna searcher with as target the mean R^2 value of the cross validation and the Optuna searcher with the penalized score introduced previously.

2.3 Sensibility to grouping

During the campaign, several characteristics differed from one point of the dataset to another one (in the way the data was taken). Those characteristics will be called label. They are representative of the spatio-temporal location of the instrument or representative of the instrument itself (for instance the id of the rabbit instruments, the campaign on which the measure was taken, the height of the sensor ...)

Consequently, the influence of grouping the data with those labels was evaluated in order to see if any improvement might appear. The data was grouped by :

- Location:
 - Column (column 1 : SL 5&11, column 2 : SL 3&4)
 - Height (155cm : SL 3&11, 270cm SL 4&5)
 - SL
- Instruments:
 - ID Rabbit
 - SL Innova (can be considered both location and instrument)
 - Couple (ID Rabbit, SL Innova)
- Campaign (June 2025 & September 2025)

2.4 Sensibility to the historic of each rabbit points

In order to try to continue improving the calibration models, several test has been performed adding engineered features on the dataframe. Those features were built thanks to the historic of each rabbit.

Let $t_{measure}$ be the time of a measurement. The idea is to introduce for each point at the time $t_{measure}$, a temporal window (of length t_{window}) of the historic of the point. We then analyse the NH3, temperature and humidity signal for t in $t_{point} - t_{window} < t < t_{point}$ (inside this window) and we add this analysis to the features for each measurement.

3 types of analysis were considered:

- Fourier analysis
- Statistical analysis
- Wavelet analysis

Several windows lengths and combinations of windows were tested. Several points are to be noted before going forward:

- The bigger the window is, the smaller the dataset is. To compare for instance the normal dataframe is composed of 8168 lines will with a window of 24h the dataset is composed (for a statistical decomposition) of 7280 lines.
- To reduce calculation time, clustering method on the features will be used meaning that only 'uncorrelated' engineered features will be used (all the base features are present). The threshold applied will be of 0.8 on pearson correlation.
- For the wavelet and fft decomposition: as those decomposition are more requiring than the statistical decomposition (and consequently less stable especially on little time windows): all the columns that have more than 15% of non valid values (compared to the regular size of the dataset) will be remove. Consequently for different time windows , the number of features increases with the window length.

2.4.1 Statistics decomposition

The following features were added: mean, std, min, max, rms, slope mean (by polyfit), last slope (slope between $t_{point-1}$ and $t_{point-2}$)

2.4.2 Wavelet decomposition

The general idea: Continuous wavelet decomposition

The wavelet decomposition is another way to analyse a signal (as the Fourier analysis) but can be considered more powerful as this analysis use functions that can play on the length of the modulating signal to analysis the signal meaning that it can better capture time variation frequencies. For instance if the mother wavelet has a length of 10mn and that a variation occurred within a second: the fourrier transform will analyse all the 10mns (as a cos function is time infinite) meaning that this variation will be "smoothed" on 10mn while with the wavelet decomposition and the ability to vary the length of the ψ we'll be able to take a window of few seconds around this variation. A general form for the wavelet is as followed (with a and b respectively frequency variation and time shifting parameters):

$$\psi_{a,b}(t) = \frac{1}{\sqrt{a}}\psi\left(\frac{t-b}{a}\right) \mid \text{continuous wavelet}$$

ψ need to be time compact and with a mean of 0.

Discrete wavelet analysis

Although powerful, the continous wavelet decomposition is a bit slow, redundant, and take much more storage space. That's why the discrete wavelet analysis is usually used.

Let X be the signal to analyze of length N. Let h_k, g_k be respectively the coefficients of discret low-pass and high-pass filter both of length F. The wavelet decomposition works as followed:

- The signal goes through different levels of filtering. At each level, the signal is processed with two filters in parallel. We collect the filters results in respectively the vector C_l (principal components of the level l) of length $N/2^l$ and the vector D_l (details of the level l) of length $N/2^l$. The details are stored and are not touched anymore while the principal components are reused for the analysis of the next level. Each coefficient of C_l ($C_{l,k}$) and each coefficient of D_l ($D_{l,k}$) is calculated as following (with $l>0$ and $C_0 = X$):

$$C_{l,k} = \sum_{j=0}^{F-1} h_j C_{l-1}[2k-j]$$

$$D_{l,k} = \sum_{j=0}^{F-1} g_j C_{l-1}[2k-j]$$

Concretely for each point k ($C_{l,k}$ or $D_{l,k}$), a sliding window is created on the previous level of principal components starting at $C_{l-1}[2k]$ and extending on the length of the filter F until $C_{l-1}[2k-(F-1)]$. The filter is then applied on this sliding window. These sliding windows are created starting every two points of the previous principal components. For the limits special conditions are applied (padding, mirror,...). The process end when the number max of level (defined by the user) is reached or when the length of the filter is superior to $N/2^l$

- At the end of the process we have C_{lmax} principal component of the signal and the details (order in decreasing importance) [$D_{lmax}, D_{lmax-1}, \dots, D_1$]

The coefficients filters h_k and g_k are directly built from strict properties arising from the imposed conditions on the mother wavelet ψ and the scaling function φ (derived from ψ) by the mathematical structures MRA (multi resolution analysis) and QMF (quadrature mirror filters)

Features added

In our case, the discrete wavelet decomposition has been used using Daubechies order 2 function and by imposing a maximum level of decomposition of. All our temporal windows can not capture the 3 levels due to the length of eadh window. Find following the table of the maximum levels by time window length:

Window length (h)	1	2	3	4	6	8	12	24
Max level	0	1	2	2	3	3	3	3

Table 2: Maximum level of wavelet decomposition per window size

We then created features from C_{lmax} :

- Energy = $\frac{1}{n_{lmax}} \sum_k^{n_{lmax}} C_{lmax,k}^2$
- Relative energy = $\frac{\frac{1}{n_{lmax}} \sum_k^{n_{lmax}} C_{lmax,k}^2}{E_{tot,signal}}$
- Root mean squared error of $C_{lmax} = \sqrt{\frac{1}{n_{tot}} \sum_{k=0}^{n_{tot}} (C_{lmax,k} - \text{mean}(C_{lmax}))^2}$

For the details level (limited to a maximum level of 3), it was created when possible for the windows for each level j:

- Energy = $\frac{1}{n_j} \sum_k^{n_j} C_{j,k}^2$
- Relative energy = $\frac{\frac{1}{n_j} \sum_k^{n_j} C_{j,k}^2}{E_{tot,signal}}$

2.4.3 Fourier

When it was possible, the first 3 peaks (maximum) on the spectrum of the temporal windows were added to the features as well as the power of each peak (defined as the squared amplitude) for a total of 6 new features per window (maximum).

3 Results

3.1 Initial results

To evaluate the performance of the instruments (and the future sections the model), the R^2 coefficient has been used. Comparing directly the NH3 concentration of the innova to the rabbit: the following scores were obtained:

R^2	MAE	RMSE
-6.65	2.59	3.8

Table 3: Initial scoring

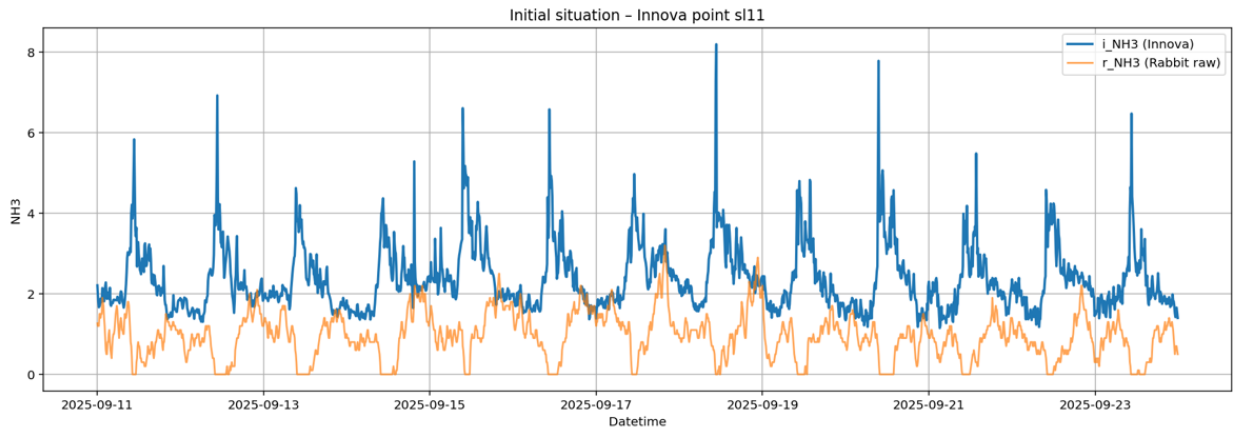


Figure 2: Initial situation

The raw IoT measurements showed limited agreement with the reference instrumentation.

The temporal evolution of the NH3 signal revealed a marked daily cyclical pattern, characterized by concentration peaks occurring approximately around midday. However, under high-concentration conditions, the IoT-based sensor exhibited a decreasing trend while the reference instrument continued to detect increasing

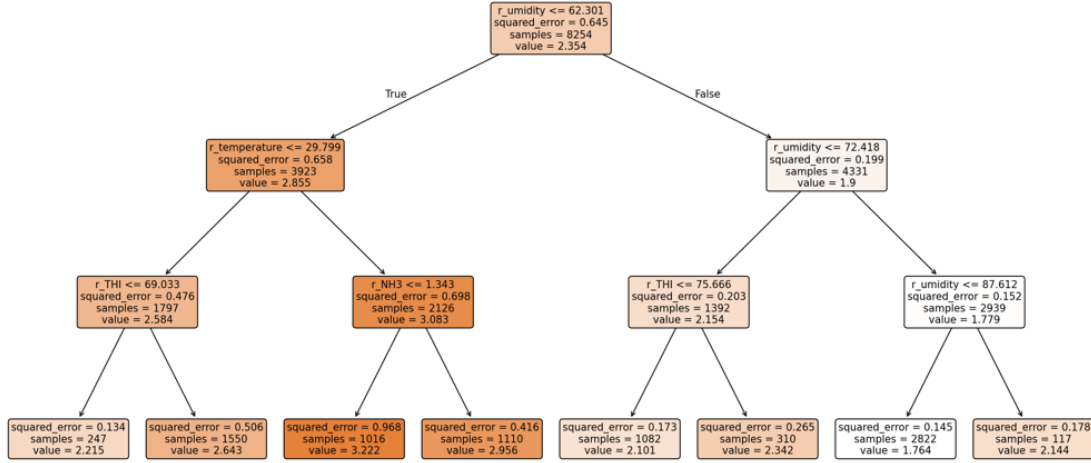


Figure 3: Decision tree of the situation

NH3 concentrations. This behaviour suggests the occurrence of sensor saturation phenomena, preventing the low-cost sensing system from accurately tracking the magnitude of the NH3 peaks measured by the reference instrumentation. Those saturation periods have a length varying between 8 and 12 hours.

TO EVALUATE IF USEFULL OR NOT

To further investigate the influence of environmental variables on sensor behaviour, a simplified Decision Tree Regression model was implemented (Figure X). The analysis highlighted relative humidity and temperature as the primary variables governing the discrepancies between the IoT-based and reference measurements for NH3 and PM2.5 . In particular, the first partitioning criterion of the tree was associated with relative humidity, followed by temperature and THI, indicating the dominant role of environmental conditions in determining sensor response. The decision tree identified distinct environmental regimes associated with different sensor behaviours, particularly for relative humidity ($RH < 62\%$, $62\% < RH < 72.5\%$, and $RH > 72.5\%$) and temperature ($T < 29.8^\circ\text{C}$ and $T > 29.8^\circ\text{C}$). These results suggest that the performance of the low-cost IoT sensor is strongly affected under specific thermo-hygrometric conditions, likely due to sensor saturation and sensitivity limitations occurring at elevated humidity and temperature levels. Consequently, the observed environmental dependence supported the implementation of correction models incorporating thermo-hygrometric variables and temporal descriptors in order to improve the predictive capability of the IoT-based monitoring system.

3.2 First tests

The standard protocol has been used on the differents models cited before. The R2 results are found on the following table:

Model	Linear	MLP	XGB	RFR	LGBM	CatBoost
R2	0.345	0.418	0.428	0.406	0.490	0.507

Table 4: Results of differents baseline models

It can be seen that the models LGBM and CatBoost are the ones performing the best. Consequently they will be kept for the remaining of the work (with also the Linear model as it is a great baseline).

3.3 Hyper-parameters sensibility and data selection

3.3.1 Hyper parameters sensibility

The linear model has no hyper parameters consequently it is impossible to do an hyper parameters sensibility, however you can still find here after the characteristics of the Linear model applied on this dataset:

Models	R2 train CV	R2 test CV	R2 test	R2trainCV/R2testCV
Linear	0.36	0.365	0.345	0.986

Table 5: Results of the Linear model

The influence of the hyper parameters of LGBM and CatBoost models were studied thanks to the Optuna searcher Optuna. Optuna searcher is an hyperparameter optimization framework that efficiently searches the best parameters using Bayesian optimization. The best and worst scores are presented here after (for two type of target: the mean R2 of the cross validation and the penealized score introduce before).

Models	R2 train CV	R2 test CV	R2 test	R2trainCV/R2testCV
Optuna MeanValCV - Best test				
LGBM	0.697	0.5635	0.6124	1.238
CatBoost	0.836	0.586	0.638	1.428
Optuna MeanValCV - Worst test				
LGBM	0.421	0.413	0.445	1.019
CatBoost	0.458	0.449	0.464	1.033

Table 6: Results of the Optuna framework with target: Mean Value Cross Validation

Models	R2 train CV	R2 test CV	R2 test	R2trainCV/R2testCV
Optuna Penalized score: $\lambda = 5$ - Best test				
LGBM	0.469	0.448	0.475	1.047
CatBoost	0.526	0.502	0.528	1.048
Optuna Penalized score: $\lambda = 5$ - Worst test				
LGBM	0.609	0.526	0.569	1.158
CatBoost	0.878	0.573	0.626	1.533

Table 7: Results of the Optuna framework with target: Penalized score

For the Mean R2 value CV target: the target is indeed optimizes as the best scores are found around a value 0.6. The gap between the best and worst test is important (around 0.2). However the gap between the R2 train CV and R2 test CV is very high and the ratio exceeds a lot of time 1.05/1.10 meaning that the model is overfitting. The model can not be seen well trained and will probably not generalize well on a whole different dataset.

For the penalized score: it can be seen that the best tests are the ones with the less gap between the value of train and test inside the CV (what was targeted). The worst results has a better R2 test than the best test however the ratio of the worst is indeed very high.

3.3.2 On the reproductibility of those results

The tests are usually done with Xtrain and Xtest separated with a random split as following (it might be seen as a perfect case):

External Split: Train: 5717 samples Test: 2451 samples

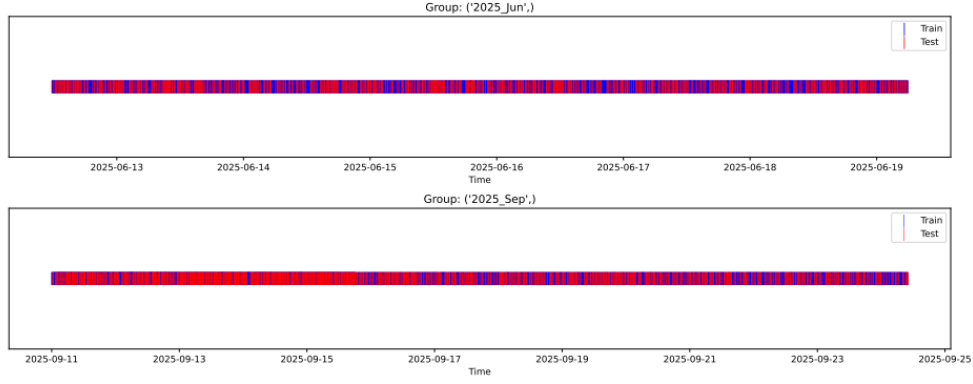


Figure 4: Random separation of Xtrain and Xtest

To test the generalisability of the model, the following separation is now tested:

External Split: Train: 5719 samples Test: 2449 samples

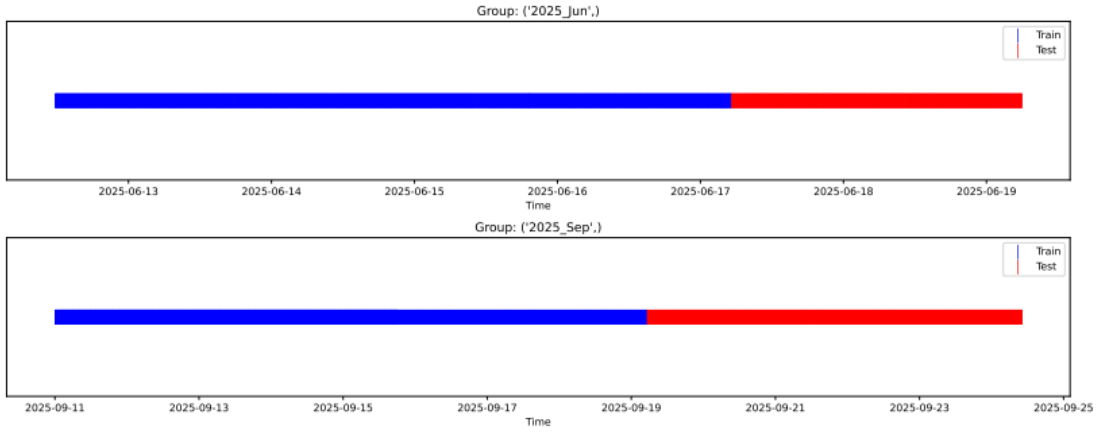


Figure 5: Time Series separation of Xtrain and Xtest

The results are presented in the following table

Models	R2 train CV	R2 test CV	R2 test	R2trainCV/R2testCV
Linear	0.403	0.408	0.187	0.987

Table 8: Results of the Linear model on the TimeSerie split

Models	R2 train CV	R2 test CV	R2 test	R2trainCV/R2testCV
Optuna MeanValCV				
LGBM	0.777	0.646	0.152	1.204
CatBoost	0.935	0.676	0.16	1.383
Optuna Penalized score: $\lambda = 5$				
LGBM	0.573	0.546	0.273	1.049
CatBoost	0.602	0.576	0.288	1.046

Table 9: Results of LGBM and CatBoost on the TimeSerie split (Mean R2 value and Penalized score

The first observation is that all the R^2 tests are reduced: the model struggle more with new data but it was expected. You can find here after a plot showing a clear degradation of the quality of the curve on the last days (where the model has not been trained).

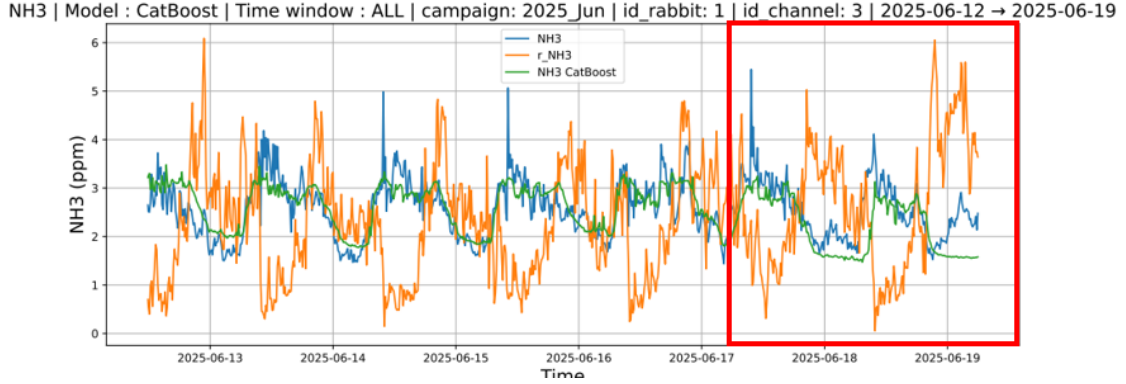


Figure 6: Model degradation plot

The second observation is that the model that have less overtraining (ie the optuna optimization with the penalized score as target) has better results.

Consequently, to try to keep it the most generalisable possible, the target will be choosen for the rest of the paper as the penalized score to prevent at maximum the overfitting that can ruin the the **FIND WORD** generalisation/ systematisation/ upscaling of the model.

The random separation is kept for the future tests (keeping in mind that real score might be lower). You can find here after

3.4 Sensibility to grouping

On this section, the impact of grouping the data by label is studied. The following table regroupes all the results as well as the % upgrade compared to the reference model defined as $(R^2 - R^2_{ref})/R^2_{ref}$.

To be noted: on the following of the paper, some tags will be add to the results to highlight (if there is) some slight problem on the training. The tags will reference to $r = R^2_{train_CV}/R^2_{test_CV}$. The tags are the following: O: $r > 1.05$; O+ $r > 1.10$; O++ $r > 1.3$; U $r < 0.95$; U- $r < 0.9$

Grouping	len group	R^2 Linear	R^2 LGBM	R^2 CatBoost
Reference				
ALL MEAN	8168	0.345	0.48	0.506
Grouped by SL				
SL 3	1356	0.641	0.64 O+	0.646 O+
SL 4	1360	0.277	0.293 O+	0.341 O+
SL 5	2726	0.457	0.459 O	0.477 O
SL 11	2726	0.394	0.574	0.578
MEAN weighted	8168	0.437 (+26.54%)	0.499 (+4%)	0.516 (+2%)
Grouped by column				
Column 1	5452	0.43	0.568	0.61
Column 2	2716	0.378 U	0.441	0.498
MEAN weighted	8168	0.423 (+22.64%)	0.526 (+9.6%)	0.572 (+13%)
Grouped by height (cm)				
155	4082	0.45	0.584	0.63
270	4086	0.364	0.349 O	0.406 O
MEAN weighted	8168	0.407 (+17.96%)	0.466 (-2.9%)	0.518 (+2.4%)
Grouped by rabbit				
Rabbit 1	795	0.5	0.536	0.582 O
Rabbit 2	2726	0.394	0.574	0.609
Rabbit 3	568	0.203 O+	0.072 O	0.187 O+++
Rabbit 4	2726	0.457	0.436 O	0.468 O
Rabbit 6	1353	0.412	0.429	0.483 O
MEAN weighted	8168	0.415 (+20.30%)	0.465 (-3.1%)	0.509 (+0%)
Grouped by (rabbit, SL)				
(1, SL3)	795	0.5	0.519	0.586
(2, SL11)	2726	0.394	0.58	0.607
(3, SL4)	568	0.203 O+	0.066 O	0.131 O+++
(4, SL5)	2726	0.457	0.439 O	0.458 O
(6, SL3)	561	0.623 U	0.702	0.703
(6, SL4)	792	0.264	0.338 O	0.455 O+
MEAN	8168	0.415 (+20.34%)	0.476 (-1%)	0.514 (+1.6%)
Grouped by campaign				
2025_Jun	3172	0.25	0.538	0.549
2025_Sep	4996	0.349	0.435	0.44
MEAN weighted	8168	0.311 (-9.98%)	0.475 (-1%)	0.482 (-4.7%)
Grouped by (campaign, column)				
('2025_Jun', 1)	1585	0.396 U-	0.599	0.669
('2025_Jun', 2)	1587	0.217	0.405 O	0.392
('2025_Sep', 1)	3867	0.45 O	0.579	0.607
('2025_Sep', 2)	1129	0.374 O	0.345 O	0.424 O+
MEAN weighted	8168	0.384 (+11.23%)	0.517 (+7.7%)	0.552 (+9.1%)

Table 10: Models results by grouping type (mean weighted)

Several points can be learnt from the those tests:

- The linear model seems to comport itself better on almost all groupings (except the campaign one): there is not a "unique" linear model for the calibration and the calibration depends of the instrument

and location.

- Regarding the decision trees models: only the location seems to improve the performance a bit (SL and column):the geometry have a bit of influence on the learning.

However as we're trying to create an universal model to be applied regardless the geometry of the infrastructure (the farm), we can consider that an upgrade of 13% on the best model (in absolute value) can be neglected. Find following a graph comparing the reference NH3 correction with the NH3 model corrected by columns backing up this assertion.

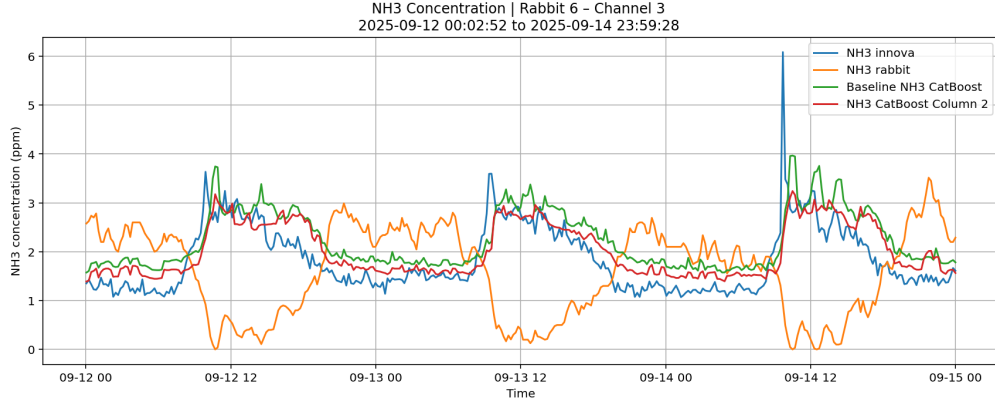


Figure 7: NH3 concentration corrected with the best grouped model

It can indeed be seen that the models differ a bit but there is not a big change on our valuation of the NH3 concentration.

3.5 Sensibility to the historic of each rabbit points

On this part, the influence of the historic of the sensor will be evaluated for the different decomposition methods as well as for different combination of time windows.

3.5.1 Statistics

The statistical decomposition is the first to be studied. Find following the table of the results for different time length as well as the % of upgrade regarding the reference model.

Window size	Len(Data)	N° of features	R^2 Linear	R^2 LGBM	R^2 CatBoost
Reference					
None			0.345	0.475	0.528
Single window					
2h	8104	15	0.537 (+55.6%)	0.694 (+46.1%)	0.696 (+31.8%)
3h	8068	14	0.507 (+47.0%)	0.68 (+43.1%)	0.669 (+26.7%)
4h	8028	14	0.546 (+58.2%)	0.629 (+32.4%)	0.636 (+20.5%)
6h	7952	14	0.483 (+40%)	0.628 (+32.4%)	0.646 (+22.3%)
8h	7876	14	0.529 (+53.3%)	0.718 (+51.2%)	0.727 (+37.7%)
12h	7730	16	0.584 (69.3%)	0.699 (47.2 %)	0.703 (+33%)
24h	7280	14	0.543 (+57.4%) O	0.639 (+34.5%) O	0.615 (+16.5%) O
Multi windows					
2h/8h	7876	24	0.58 (+68.1%)	0.728 (+53.3%)	0.741 (+40.3%)
2h/12h	7730	26	0.591 (+71.3%)	0.654 (+37.7%)	0.67 (+26.9%)
8h/12h	7730	25	0.588 (+70.4%)	0.685 (+44.2%)	0.7 (+36.4%)
2h/3h/4h	8028	33	0.578 (+67.5%)	0.603 (+26.7%)	0.648 (+22.7%)
8h/12h/24h	7280	34	0.616 (+78.5%) O+	0.692 (+45.7%) O+	0.583 (+10.4%) O+
2h/8h/12h	7730	35	0.599 (+73.2%)	0.671 (+41.3%)	0.669 (+26.7%)

Table 11: Results of the statistical decomposition

Adding features of the historic of the rabbit have a significant and positive impact on the results of the calibration of all the cases.

It appears also that the choosing of the time window as an impact on the performance of this upgrade. For the single window part we can distinct several performances:

- The 24h window seems to underperform. It might be explained first by a slight overtraining residual combined to the fact that the 24h time window remove more points of the training (ie length total of the data set of 7280 vs 8168 for the baseline dataset). Another explanation might be that the time window is too big to reflect any internal behaviour of the sensor.
- The windows 2h, 8h and 12h are the ones that seems to overperform. The overperformance of the 8 and 12h window might be explained because of the time period of visible saturation of the instrument (as explained in the initial results) as it varies between more or less 8/12h. The overperformance of the one of 2h might be explained by the fact that the window is enough big to have a real statistical value but not too large to explain the "fast" variations.
- Combining the windows can be a good idea but to use carefully. Indeed the linear model seems to improve no matter what combination (even if we include the slight overtraining). However for the 2 other models, adding more and more information do not help the model learning (the test with only 8h and the test with the windows 2h/8h have better results than all the other tests with combined windows).

3.5.2 Wavelet

The next decomposition is the wavelet one. Find following the table of the results for different time length as well as the % of upgrade regarding the reference model.

Window size	Len(data)	N° of features	R^2 Linear	R^2 LGBM	R^2 CatBoost
Reference					
None	8168	5	0.345	0.475	0.528
Simple window					
2h	8033	15	0.495 (+43.5%)	0.672 (+41.5%)	0.670 (+26.9%)
3h	8032	16	0.562 (+62.9%)	0.62 (+26.3%)	0.653 (+23.7%)
4h	8010	17	0.509 (+47.5%)	0.588 (+23.8%)	0.632 (+20.1%)
6h	7932	19	0.537 (+55.6%) O+	0.636 (+34.0%) O	0.636 (+20.5%) O
8h	7852	20	0.528 (+53.0%)	0.707 (+48.8%)	0.709 (+34.3%)
12h	7698	20	0.508 (+47.2%)	0.671 (+41.3%)	0.702 (+33.0%)
24h	7224	15	0.387 (+12.2%)	0.565 (+19.0%) O	0.493 (-6.6%)
Multi window					
2h/8h	7793	30	0.574 (+66.4%) U	0.758 (+59.6%)	0.766 (+45.1%)
2h/12h	7639	28	0.56 (+62.3%)	0.71 (+49.5%)	0.714 (+35.2%)
8h/12h	7698	26	0.55 (+59.4%)	0.657 (+38.3%)	0.683 (+29.4%)
2h/3h/4h	7953	38	0.543 (+57.4%)	0.565 (+19.0%)	0.598 (+13.3%) O
8h/12h/24h	7224	44	0.568 (+64.6%)	0.643 (+35.4%)	0.636 (+20.5%)
2h/8h/12h	7639	46	0.599 (+73.6%)	0.705 (+48.4%)	0.714 (+35.2%)

Table 12: Results of the statistical decomposition

The interpretation is the same as the statistical decomposition: the windows 8 and 2h/8h are the one performing the best. In absolute value the best case scenario is slightly better for the wavelet decomposition 2h/8h (0.766 vs 0.741 for the statistical).

3.5.3 Fourrier

The last decomposition is the fourrier one. Find following the table of the results for differents time length as well as the % of upgrade regarding the reference model.

Window size	Len(data)	N° of features	R^2 Linear	R^2 LGBM	R^2 CatBoost
Reference					
None	8168	5	0.345	0.475	0.528
Simple window					
1h					
2h	8092	11	0.479 (+38.8%)	0.58 (+22.1%)	0.582 (+10.2%)
3h	8052	11	0.463 (+34.2%)	0.546 (+15.0%)	0.563 (+6.6%)
4h	8012	11	0.4 (+15.9%)	0.538 (+13.3%) O	0.513 (-2.8%)
6h	6821	16	0.37 (+7.2%)	0.423 (-10.9%)	0.435 (-17.6%)
8h	7072	17	0.338 (-2.0%)	0.468 (-1.5%)	0.454 (-14.0%)
12h	7652	21	0.41 (+18.8%)	0.555 (+16.8%)	0.547 (+3.6%)
24h	7224	22	0.375 (+8.7%)	0.439 (-7.6%)	0.451 (-14.6%)

Table 13: Results of the fourrier decomposition

It was chosen not to proceed with the multi windows as the results were less much encouraging than the 2 others decompositions.

3.6 Additional results

To ensure that this method is usable in real conditions, it was decided to try the feature decomposition (wavelet 2h/8h) with a time serie split (as in the reproductibility part). To avoid data leakage a 8h gap between the Xtrain and Xtest was added as followed:

You can find following the results of those tests in a table as well as a plot showing the improvement:

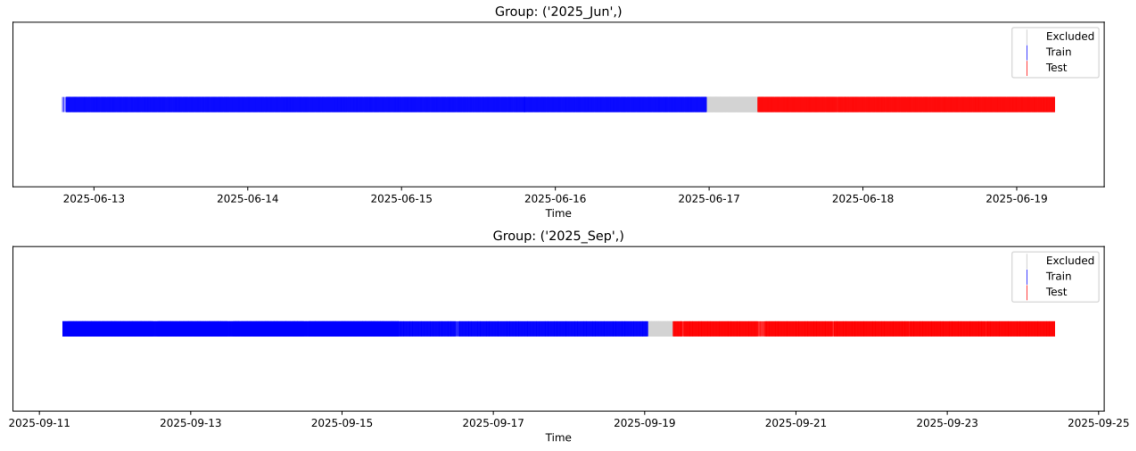


Figure 8: Xtrain and Xtest repartition with an 8h gap

Window size	R^2 Linear	R^2 LGBM	R^2 CatBoost
TimeSeriesSplit Reference			
None	0.187	0.273	0.288
TimeSeriesSplit Wavelet decomposition			
2h/8h	0.304	0.508	0.512

Table 14: Results of the decomposition on a timeserie split

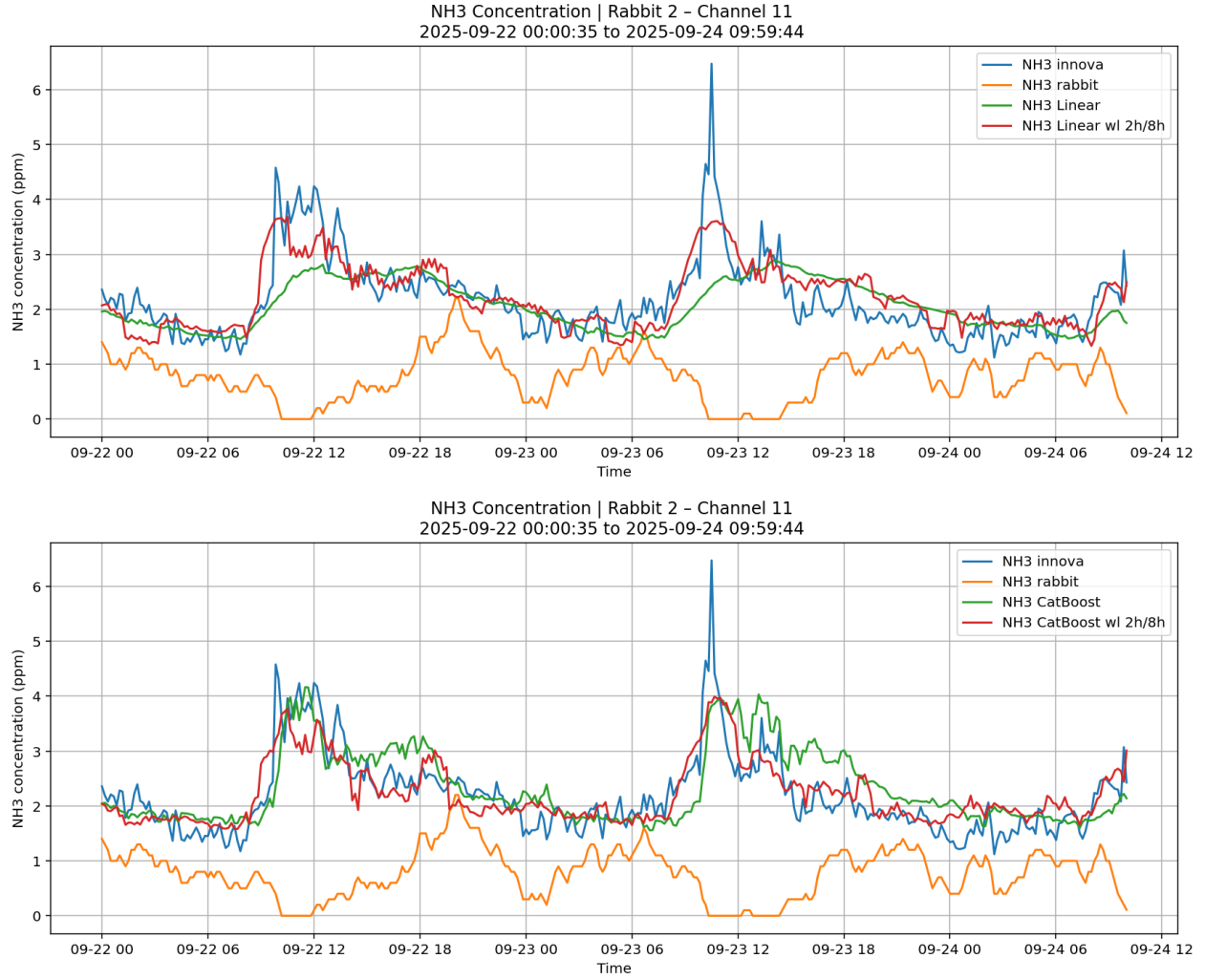


Figure 9: Plot results of the decomposition on a time serie split

In conclusion, it can be said that whether we take or no the time serie split the model performs better with the engineered features.

4 Conclusion and discussion

This paper focused on the sensibility parameters of the calibration of a low cost sensor for NH3 on a high cost one. Several points are to be noted:

- The machine learning models used in this paper (LGBM, CatBoost) can overfit quickly. It is of utmost importance to regulate the learning in order to be able to deploy the calibration at large scales.
- The geometry of the farm might have an influence on the calibration process, however it can be neglected at first order.
- Adding features from the historic of the low cost sensor can greatly help to enhance the precision of the calibration. For the simplicity of implementation a statistical implementation might be used on the combination of windows of 2h and 8h.

Following this paper, several points are still to be deepen:

Further studies might look at other ways to decompose the historic in order better again the calibration (especially other wavelet filters might be used).

More importantly: it would be usefull to confirm the previous conclusions on another gas, on other instruments, or in other geometry configurations. Further studies can also focus on the optimization of the time length regarding the instrument, the gas (...) in order to understand which elements to take into account to calibrate at best our low cost instruments. Finally an interesting test would be to check whether or not the model trained during this work might be reusable for others on different (or whether it needs to be retrained i.e **PAPER citation** configurations to understand at which point this model is yet scalable.

Further studies might focus on other ways to decompose the signal in order to find the best model possible. The wavelet decomposition can also be furthered as the study was done with really basic parameters. Finally we can also think of combining all those decompositions in order to see if one model can help another one.